THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. All the 12 faces of a polyhedron are pentagons. Its number of vertices is: 1

(a) 20 **(b)** 18 **(c)** 44 **(d)** 30

2. The mean of 5 numbers is 30. When one number is removed the mean of the remaining 4 becomes 27. The removed number is: 1

(a) 3 **(b)** 15 **(c)** 30 **(d)** 42

3. A two-digit number exceeds the number formed by reversing its digits by 18. The difference of its digits is: 1

(a) 3 **(b)** 2 **(c)** 1 **(d)** 18

4. The perimeter of a rectangle is 128 cm and the length is 3 times the breadth. Its area is: 1

(a) 1024 cm² **(b)** 784 cm² **(c)** 736 cm² **(d)** 768 cm²

5. A polyhedron has 18 edges and exactly 3 edges meet at every vertex. Its number of faces is: 1

(a) 8 **(b)** 12 **(c)** 20 **(d)** 6

6. The mean of 10 values was found to be 36. Later it was found that 80 was misread as 30. The correct mean is: 1

(a) 86 **(b)** 36 **(c)** 41 **(d)** 31

7. If $x + \frac{1}{x} = 3$, then $x^2 + \frac{1}{x^2}$ equals: 1

(a) 7 **(b)** 11 **(c)** 6 **(d)** 9

8. An isosceles trapezium has parallel sides 10 cm and 22 cm and non-parallel sides 10 cm. Its area is: 1

(a) 128 cm² **(b)** 160 cm² **(c)** 138 cm² **(d)** 256 cm²

9. If every corner of a cube is sliced off (without removing the whole edges), the number of vertices of the resulting solid is: 1

(a) 16 **(b)** 12 **(c)** 24 **(d)** 8

10. A class has 29 students with mean age 13 years. When the teacher's age is included, the mean rises to 14. The teacher's age (years) is: 1

(a) 38 **(b)** 42 **(c)** 43 **(d)** 14

11. The sum of the dates in a 2×2 calendar block is 44. The smallest date in the block is: 1

(a) 6 **(b)** 8 **(c)** 15 **(d)** 7

12. The length of a rectangle is increased by 20% and its breadth decreased by 10%. If the original area is 1000 cm², the new area (in cm²) is: 1

(a) 1000 **(b)** 1100 **(c)** 1080 **(d)** 1200

13. **Assertion (A):** A cube satisfies Euler's formula. 1

Reason (R): For every convex polyhedron, $F + V - E = 2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** The mean of 4, 6, 8, 10 and 12 is 8. 1

Reason (R): The mean of a data set is the sum of the observations divided by the number of observations.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** For any whole number n, the number $2n$ is even and $2n + 1$ is odd. 1

Reason (R): Even numbers are multiples of 2, and odd numbers leave remainder 1 when divided by 2.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A cube is painted all over and cut into unit cubes. If 48 unit cubes have exactly two painted faces, find the side of the cube and the total number of unit cubes. 2

17. The mean of 6 numbers is 20. When one number is removed, the mean of the remaining 5 numbers is 18. Find the removed number. 2

18. The sum of the digits of a two-digit number is 9 and the tens digit exceeds the ones digit by 3. Find the number. 2

19. The perimeter of a rectangle is 100 cm and its length exceeds its breadth by 18 cm. Find its area. 2

20. A polyhedron has 8 triangular faces and 6 square faces only. Find its number of edges. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Every face of a polyhedron is a triangle and it has 32 faces. Find the number of edges and vertices and verify Euler's formula. 3

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22. A class of 40 students has mean mark 62. After rechecking, a score of 38 was corrected to 80. Find the new mean. 3

23. The sum of a two-digit number and its reverse is 110 and the digits differ by 4. Find the number (tens digit larger). 3

24. The perimeter of a rectangular plot is 160 m and its length is 3 times its breadth. Find its area and the cost of fencing at ₹ 15 per metre. 3

OR

A cube is painted all over and cut into unit cubes. For which side is the number of unit cubes with exactly two painted faces equal to 144? Hence find the number with exactly one painted face. 3

25. The mean age of 11 players is 23 years. When the coach is included the mean becomes 25. Find the age of the coach. 3

OR

If $x + \frac{1}{x} = 3$, find $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Rohan makes solid models of a pentagonal prism and a pentagonal pyramid for a school exhibition and counts their faces, vertices and edges. 5

(i) How many faces does the pentagonal prism have?

(ii) How many vertices does it have?

(iii) How many edges does it have?

(iv) Verify Euler's formula for the prism and for the pentagonal pyramid.

27. Prove that $(2n+3)^2 - (2n-3)^2 = 24n$ and use it to evaluate it for $n = 4$. 5

OR

The area of a rhombus is 216 cm² and one diagonal is 18 cm. Find the other diagonal, the side, and the altitude. 5

END OF QUESTION PAPER — Total: 50 Marks

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